

String functions

Where to find the practice data

- We're working with a sample of DC 311 service requests: the calls and online tickets residents file for missed trash pickup, potholes, rodent inspections, and the like.
- Make your own copy: File → Make a copy, then work in that.

PRACTICE FILE

dc311sample.csv

<https://go.umd.edu/di-string>

Creating a unique ID

- This data already ships with an ID: the SERVICEREQUESTID column, like 26-00183801. Handy, but should you trust it?
- Sheets can also auto-fill your own sequential ID, one number per observation, with no repeats:
- Insert a column to the left; name it something useful like unique_ID.
- Type 1 in the first cell. In the next, type = then click the cell above, then +1, and hit enter.
- Click the lower-right corner and drag it down the whole column.

fx **=A2+1**

2

auto-fill extends 1, 2, 3, 4 ... down

Reporting note

Every 311 ID here starts with 26- — the agency's prefix, not yours. If you merge years of data later, ask whether these IDs stay unique across files.

Changing the format of a field

- Sheets often guesses the wrong type for a column. Take a look at ADDDATE — when the request was filed.
- It arrives as one long text string: a date, a time, and a +00 on the end. That tail is the time zone.
- If you only need the calendar date, peel off the first 10 characters into a new column.
- Or select the column, open the 123 → menu, and set the format you actually want.

fx **=LEFT (M2 , 10)**

2026/03/25

from 2026/03/25 04:00:18+00

Reporting note

The +00 means these times are in UTC, not DC local time. If you ever compute “how long did a request take,” that offset shapes the answer.

Splitting cells

- Sometimes one column holds several things at once. Sheets has a handy — if imperfect — way to break it apart.
- Look at STREETADDRESS. What's bundled together in there?
- A house number, a street name, and a quadrant (NE / NW / SE / SW) - three things in one cell.
- To group by quadrant or sort by street, we'll need them in separate columns.

5224 AMES STREET NE

STREETADDRESS

5224 + AMES STREET + NE

Splitting cells

- First, add 4 empty columns to the right of STREETADDRESS. Sheets will overwrite whatever's next to it, so err on the side of too many empties.
- Select the column, then: Data → Split text to columns.
- Down at the bottom, choose Space as the separator.
- Watch what lands in each new column.

DATA ▸

Sort range...

Split text to columns

Pivot table...

Named ranges...

Separator: () Comma (•) Space

Splitting cells

Scroll through what you got. Some rows split cleanly. Others don't:

CLEAN

1000 F STREET NW

1000 | F | STREET | NW

MESSY

1300 - 1399 BLOCK OF KENNEDY ST NW

1300 | - | 1399 | BLOCK | OF | KENNEDY ...

- Same lesson as messy names: the number of pieces isn't always the same, so columns don't line up. You might need to do additional work to extract clean first and last names, or address components.

Pop quiz: what does CONCATENATE do?

Splitting pulls a cell apart. CONCATENATE does the opposite — it glues pieces back together. (It's fine to Google it.)

```
fx =CONCATENATE (H2, " — ", X2)
```

Rodent Inspection and Treatment — Ward 5

Glue the service description to its ward to make one readable label.

LEFT / RIGHT / MID

- When you know how many characters you want from a fixed spot in a cell, LEFT, RIGHT and MID grab them directly.
- LEFT takes characters from the start, RIGHT from the end, MID from the middle.
- One example is a 3-digit ZIP. Even with full ZIP codes, the first 3 digits help when data is thin or you're protecting privacy.

LEFT

→ 200

=LEFT(zip,3)

RIGHT

→ NE

=RIGHT(addr,2)

MID

→ 0018

=MID(id,4,4)

LEFT / RIGHT / MID

- The 3-digit ZIP: select the ZIPCODE column, insert a new column to its left, and name it ZIP3.
- In the first data cell, enter the LEFT formula and double-click to extend it down.
- Now the quadrant: RIGHT grabs the last 2 characters of STREETADDRESS — a clean way to get NE / NW / SE / SW without splitting.

fx **=LEFT (G2 , 3)**

200

first 3 digits of 20019

fx **=RIGHT (D2 , 2)**

NE

quadrant from the address

Cleaning text: TRIM & PROPER

- The addresses here are ALL CAPS. If you're matching or displaying them, PROPER re-capitalizes to Title Case.
- TRIM removes stray leading, trailing, and double spaces, the invisible gremlins that break sorts.
- These two are your everyday cleanup workhorses; reach for them before matching one dataset to another.

```
fx =PROPER(D2)
```

5224 Ames Street Ne

Title Case (you'll still fix "Ne")

```
fx =TRIM(" Ward 5 ")
```

Ward 5

stray spaces removed

SUBSTITUTE: find & replace in a cell

- SUBSTITUTE swaps one piece of text for another, inside a single cell — great for stripping junk.
- GLOBALID comes wrapped in curly braces: {4DD793B5-...}. Swap each brace for nothing to clean it.
- Same trick drops the word “Ward ” so you're left with just the number, handy for sorting wards in order.

```
fx =SUBSTITUTE (W2,"Ward", "")
```

5

from “Ward 5”

```
fx =SUBSTITUTE (G2," {", "")
```

4DD793B5-...

first brace gone; repeat for }

Keep exploring

Pick another text column - SERVICECODEDESCRIPTION, ORGANIZATIONACRONYM, the dates - and ask what each function could pull out, clean, or rebuild.